

A PROJECT ABSTRACT ON

COVID-19 DATA VISUALIZATION

(Data Analytics & Business Intelligence)

Project Domain	Data Analytics & Business Intelligence
Project Type	Data Visualization and Interactive Dashboard
Department	Computer Science and Engineering (Artificial Intelligence and Machine Learning)
Primary Objective	Analysis and visual representation of COVID-19 data

Submitted in partial fulfilment of academic project requirements

Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Abstract

The COVID-19 pandemic generated an unprecedented amount of public health data, including confirmed cases, recoveries, deaths, testing information, vaccination statistics, and geographical distributions. However, raw COVID-19 datasets are generally presented in tabular form, making it difficult for users to quickly understand the scale, progression, and regional impact of the pandemic. This project, titled **COVID-19 Data Visualization**, focuses on transforming raw COVID-19 data into meaningful visual insights through a structured data analytics and Business Intelligence approach. The main purpose of the project is to provide an easy-to-understand analytical view of pandemic-related information so that users can identify trends, compare regions, observe changes over time, and understand important indicators without manually examining large datasets.

The project addresses the problem of interpreting large and continuously changing COVID-19 datasets. A simple spreadsheet containing thousands of records does not immediately communicate important patterns such as increases or decreases in cases, differences between countries or regions, changes in mortality, or the relationship between cases, recoveries, and deaths. Manual analysis is also time-consuming and may make comparison difficult. Therefore, the proposed system provides a centralized visualization environment in which cleaned COVID-19 data can be explored through charts, graphs, summary indicators, and interactive filters. The project is intended as an academic-scale Data Analytics and Business Intelligence solution and focuses on descriptive analysis and visualization rather than medical diagnosis or prediction.

The project follows a structured data-processing workflow similar to the BI pipeline presented in the reference project report. The process begins with **data collection**, where a suitable COVID-19 dataset containing date-wise and location-wise pandemic statistics is obtained in CSV or spreadsheet format. The next stage is **data understanding and cleaning**, where the dataset is inspected for missing values, duplicate records, inconsistent names, invalid data types, and formatting problems. The cleaned data is then transformed into an analysis-ready structure. Date fields can be converted into day, month, quarter, and year attributes, while derived indicators such as active cases or recovery/death percentages can be calculated when the required fields are available. The prepared dataset is finally connected to the visualization layer for analysis and reporting.

The project is designed for the **Computer Science and Engineering (Artificial Intelligence and Machine Learning)** department and applies relevant concepts from data analytics, machine learning-oriented data processing, and Business Intelligence. The major tools and applications used in the project include **Python, Pandas, Microsoft Excel/CSV**, and **Power BI** or an equivalent Business Intelligence visualization platform. Python and Pandas are used for data loading, inspection, cleaning, transformation, and exploratory analysis. Excel or CSV acts as the source and intermediate data format for reviewing the dataset. Power BI is used to build interactive visualizations, dashboard pages, filters, cards, and analytical charts. Where required, DAX measures can be used to calculate important metrics inside the BI platform.

From an AIML perspective, the project also establishes a suitable foundation for future intelligent analysis. The cleaned and structured dataset can later be used for predictive modeling, time-series forecasting, anomaly detection, or other machine learning techniques. However, the present project concentrates on data visualization and descriptive analytics so that the available COVID-19 information can first be understood clearly before applying advanced predictive models. This separation ensures that the visualization layer communicates the observed data accurately while leaving machine learning extensions for future

development.

Problem Statement and Overview

The central problem addressed by the project is the difficulty of obtaining meaningful information from large COVID-19 datasets using raw tables alone. Users may need to compare total confirmed cases, deaths, recoveries, active cases, and other available indicators across countries, states, or time periods. Without visualization, identifying trends and regional differences requires repeated manual calculations. The proposed COVID-19 Data Visualization system solves this problem by organizing the data into a consistent analytical format and presenting the results through interactive visual elements. The dashboard is designed to provide a high-level overview first and then allow users to explore detailed information using filters such as date, country/region, and other available dimensions.

Sub Modules

- **1. Data Collection Module:** Imports the selected COVID-19 dataset and identifies relevant fields such as date, location, confirmed cases, deaths, recoveries, testing, vaccination, or other available indicators.
- **2. Data Cleaning Module:** Detects missing values, duplicates, inconsistent location names, incorrect data types, and formatting issues. Appropriate cleaning operations are applied before analysis.
- **3. Data Transformation Module:** Converts the cleaned dataset into an analysis-ready format and creates useful date attributes and derived measures wherever the source data supports them.
- **4. Data Analysis Module:** Calculates summary statistics and key indicators to understand case growth, mortality, recovery, regional distribution, and time-based changes.
- **5. Visualization Module:** Represents the analyzed information using KPI cards, line charts, bar charts, area charts, maps where geographic fields are available, and other suitable visuals.
- **6. Interactive Dashboard Module:** Provides filters and slicers so users can select a date range, location, or other dimensions and observe corresponding changes in the dashboard.
- **7. Insight and Reporting Module:** Presents important observations obtained from the visualizations in a clear format suitable for academic reporting and decision-support analysis.

Design / Flow of the Project



The project follows a linear analytical pipeline in which each stage produces validated input for the next stage. Raw data is first checked and cleaned, then transformed into a structured dataset. Analytical measures are calculated from the prepared data and passed to the visualization layer. The final dashboard combines summary KPIs with trend and comparison charts, enabling users to move from an overall view to more detailed observations.

Major Dashboard Components

- **KPI Summary Cards:** Display important overall indicators such as confirmed cases, deaths, recoveries, active cases, or vaccination-related measures when available in the dataset.
- **Time-Series Trend Analysis:** Uses line or area charts to show how COVID-19 indicators change across days, months, or other time periods.
- **Regional/Country Comparison:** Uses bar charts or maps to compare the pandemic situation across geographical locations.

- **Case and Outcome Analysis:** Compares confirmed cases with deaths, recoveries, and active cases to provide a clearer view of outcomes.
- **Interactive Filters:** Allows users to select specific dates, countries, states, or other dataset dimensions and update the visualizations dynamically.

Expected Output and Conclusion

The expected output of the project is an interactive COVID-19 Data Visualization dashboard that converts complex pandemic data into simple, readable, and useful visual information. The dashboard should enable users to understand the overall status of COVID-19 data, observe trends over time, compare affected regions, and examine major indicators through interactive filtering. Instead of relying only on raw tables, users can obtain a visual summary and then investigate specific parts of the dataset based on their requirements. The project therefore demonstrates the practical application of data cleaning, transformation, exploratory analysis, KPI calculation, and Business Intelligence visualization in a real-world public-health data context. Overall, the COVID-19 Data Visualization project provides a structured approach for analyzing and communicating pandemic-related data within the CSE (Artificial Intelligence and Machine Learning) domain. By combining Python/Pandas-based data preparation with Power BI-based interactive visualization, the project aims to improve data accessibility, reduce manual analysis effort, and make important patterns easier to recognize. The final system is expected to serve as an academic demonstration of how data analytics and BI tools can transform large datasets into meaningful visual insights. Future extensions may include automated data refresh, integration of additional public-health datasets, predictive trend analysis, anomaly detection, vaccination-focused analysis, and deployment of the dashboard through a cloud-based BI environment. The project does not claim medical diagnosis or clinical decision-making; its primary purpose is descriptive data analysis and visualization, with a clear pathway toward future AIML-based predictive extensions.

Key Terms

COVID-19, Data Visualization, Business Intelligence, Data Analytics, Artificial Intelligence, Machine Learning, Power BI, Python, Pandas, Dashboard, KPI, Data Cleaning, Trend Analysis, Interactive Visualization